

IRM Final Report

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Case: Student Exercise Scenario
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Date: 12/10/2025

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EXECUTIVE SUMMARY

Between June 17 and June 25, 2025, two internal Windows hosts (ACC-WIN10-13 and ACC-WIN10-16) were compromised by a malicious executable identified as **QRCodeScanner.exe**. The binary was delivered through an external domain and executed under multiple user contexts, initiating outbound command-and-control (C2) communication with a remote server accessible at **saltybug.com**.

During the incident window, both systems exhibited:

- Repeated **HTTP GET and POST communication** with the C2 server
- Automated exfiltration of user data via PowerShell
- Evidence of **credentialed execution, persistence mechanisms, and anti-forensic cleanup**
- File deletion and artifact removal to obscure attacker actions
- **Multi-host compromise**, indicating lateral movement or a common infection vector

Incident scope confirms that execution, persistence, and exfiltration behavior occurred across both affected hosts, reflecting an advanced and automated intrusion. The coordinated activity was validated through Sysmon logs, proxy logs, IDS, Registry analysis, Prefetch artifacts, Shellbag interactions, MFT metadata, and packet capture review.

A full timeline of system-level activity, forensic artifacts, and network communication was constructed to align execution events, exfiltration behavior, and forensic indicators across both systems. These findings confirm an active intruder presence, targeted data collection, and structured exfiltration during June 25, 2025.

SUMMARY

- **Incident Description:** Two internal Windows systems were compromised after executing a malicious binary named *QRCodeScanner.exe*.
- **Date and Time of Discovery:** June 25th, 2025.
- **Initial Detection:** The initial detection occurred on June 25th, 2025, when suspicious POST and GET activity from **ACC-WIN10-16** to **saltybug.com/update** was observed in proxy logs, followed by validation from Sysmon, IDS, Registry, and packet capture artifacts.
- **Affected Systems/Data:** ACC-WIN10-13, ACC-WIN10-16.
- **Estimated Impact:** The incident resulted in unauthorized remote access, automated data exfiltration, and execution of anti-forensic cleanup actions. If conducted against a production environment, this activity would pose financial, operational, and reputational risk due to exposure of internal data and sustained attacker presence.

TECHNICAL FINDINGS

Initial Compromise and Malware Delivery

Proxy logs show that ACC-WIN10-16 initiated a **HTTP GET request** to retrieve QRCodeScanner.exe from a remote host. User-agent strings and port usage indicate user-driven execution (e.g., browser-based click), consistent with malware delivery through user interaction.

14201	2025-06-25 11:19:31.723170	172.16.6.116	172.16.2.6	HTTP	192 GET http://sodium.ca/QRCodeScanner.exe HTTP/1.1
23487	2025-06-25 11:19:38.324800	172.16.6.116	172.16.2.6	HTTP	311 GET http://saltybug.com/register/acc-win10-16/windows HTTP/1.1
23509	2025-06-25 11:19:38.367342	172.16.6.116	172.16.2.6	HTTP	290 GET http://saltybug.com/update/1 HTTP/1.1
23577	2025-06-25 11:20:12.400561	172.16.6.116	172.16.2.6	HTTP	290 GET http://saltybug.com/update/1 HTTP/1.1
23615	2025-06-25 11:20:16.460753	172.16.6.116	172.16.2.6	HTTP	403 POST http://saltybug.com/update HTTP/1.1 (text/plain)
23635	2025-06-25 11:20:16.538719	172.16.6.116	172.16.2.6	HTTP	311 GET http://saltybug.com/register/acc-win10-16/windows HTTP/1.1

Prefetch artifacts and MFTECmd results confirm the executable was written to disk and executed immediately following download. Creation, modification, and run counts validate repeated use.

Execution and Credentialed Elevation

Sysmon Event ID 1 confirms execution by multiple user contexts, including administrative elevation and credential passing via PowerShell. Parent processes show that PowerShell launched QRCodeScanner.exe with execution privileges and hidden window style.

Relevant actions include:

- Credential object creation in PowerShell
- Elevated execution of malware using `runas -windowstyle hidden`
- PowerShell loop processing local files and automatically exfiltrating random content

s/25/2025	11:20:17 AM	-	ACC-WIN10-16 (172.16.6.116)	Proxy Logs	First HTTP POST to C2/exfil domain saltybug.com (162.241.253.54)	Splunk-site_proxy	FI
s/25/2025	11:20:17 AM	-	acc-win10-16.site.lan	Windows Logs	400,"68527,68527,2025-06-25 11:20:17.2310769,400,info,PowerShell,Windows PowerShell,0,0,acc-	Splunk-windows_logs	PI
s/25/2025	11:20:17 AM	-	ACC-WIN10-16 (172.16.6.116)	Proxy Logs	src_ip=172.16.6.116 src_port=60493 dest_ip=162.241.253.54 dest_port=80 http_method=GET url="h	Splunk-site_proxy	BI
s/25/2025	11:20:17 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Users\Public\QRCodeScanner.exe Commandl	Splunk-windows_sysmon	Q
s/25/2025	11:20:17 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - QRCodeScanner.exe executed with elevated privileges	Splunk-windows_sysmon	Q
s/25/2025	11:20:17 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - QRCodeScanner.exe executed with elevated privileges by PowerShell	Splunk-windows_sysmon	BI
s/25/2025	11:20:17 AM	-	ACC-WIN10-16 (172.16.6.116)	Prefetch	Prefetch entry for C:\Users\Public\QRCodeScanner.exe shows LAST run time 2025-06-25 11:20:17	PECmd - Prefetch artifacts	LI

Proxy logs and PCAP review confirm **C2 registration via saltybug.com/update** shortly after execution.

Persistence and Scheduled Task Creation

Sysmon and Registry confirm creation and execution of a malicious scheduled task:

- Task Name: **ThreadCleaner**
- Execution target: **C:\Windows\vss\vixDiskMountServer.exe**
- Task registered under SYSTEM context
- Scheduled at logon for ongoing C2 control or cleanup

This pattern is consistent with attacker persistence.

6/25/2025	11:38:16 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Windows Logs	Event ID: 4648 LogonAttemptExplicitCredentials TargetServerName acc-win10-13 IpAddress 172.17.1 Splunk-windows_logs	Continued credential usage attempts between hosts
6/25/2025	11:38:52 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\cmd.exe CommandLine c:\splunk-windows_sysmon	Scheduled task creation for persistence
6/25/2025	11:38:52 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Windows Logs	Event ID: 4648 LogonAttemptExplicitCredentials TargetServerName acc-win10-13 TargetUser SITE splunk-windows_logs	Another explicit credential attempt via schtasks.
6/25/2025	11:38:52 AM	SITE\LAN\Administ	acc-win10-13.site.lan	Windows Logs	Event ID: 4624 SuccessfulLogon LogonType 3 AuthenticationPackageName Kerberos IpAddress 172.17.1 Splunk-windows_logs	Kerberos authentication success from acc-win10-16
6/25/2025	11:38:52 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - cmd.exe executed schtasks.exe to create 'ThreadCleaner' task on ACC-WIN10-16 Splunk-windows_sysmon	Persistence mechanism created via scheduled task
6/25/2025	11:38:52 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - schtasks.exe created task 'ThreadCleaner' to run QRCodeScanner.exe Splunk-windows_sysmon	Scheduled task established to execute QRCodeScanner.exe
6/25/2025	11:39:25 AM	-	acc-win10-16.site.lan	Sysmon Logs	Scheduled Task created: ThreadCleaner Splunk-windows_sysmon	Persistence mechanism
6/25/2025	11:39:25 AM	-	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	src_ip=172.16.6.113 src_port=60883 dest_ip=162.241.253.54 dest_port=80 http_method=GET url=https://splunk-site_proxy	Second host shows C2 communication, indicating initial exfiltration from compromised host
6/25/2025	11:39:25 AM	SYSTEM	ACC-WIN10-16 (172.16.6.116)	Proxy Logs	POST request to http://saltybug.com/update initiated site_proxy	Scheduled task authentication attempt to acc-win10-13
6/25/2025	11:39:26 AM	ACC-WIN10-16\Adm	acc-win10-16.site.lan	Windows Logs	Event ID: 4648 LogonAttemptExplicitCredentials TargetServerName acc-win10-13 TargetUser SITE splunk-windows_logs	Successful Kerberos network logon from acc-win10-16
6/25/2025	11:39:26 AM	SITE\LAN\Administ	acc-win10-13.site.lan	Windows Logs	Event ID: 4624 SuccessfulLogon LogonType 3 AuthenticationPackageName Kerberos IpAddress 172.17.1 Splunk-windows_logs	Exfiltration begins from second compromised host
6/25/2025	11:40:00 AM	SYSTEM	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	POST request to http://saltybug.com/update initiated site_proxy	File copied for persistence
6/25/2025	11:40:00 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\cmd.exe CommandLine c:\splunk-windows_sysmon	QRCodeScanner copied for persistence or privilege escalation
6/25/2025	11:40:00 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - cmd.exe copied QRCodeScanner.exe to C:\Windows\Vs\viaDiskMou Splunk-windows_sysmon	Sensitive file collection
6/25/2025	11:40:40 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	PowerShell commands reference files in C:\Windows\vs\secret1.txt, secret2.txt Splunk-windows_sysmon	Persistence via scheduled task
6/25/2025	11:40:40 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\WindowsPowerShell\cmd.exe CommandLine c:\splunk-windows_sysmon	Persistence established using PowerShell scheduled task
6/25/2025	11:40:40 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - PowerShell created scheduled task 'ThreadCleaner' executing vixDisk Splunk-windows_sysmon	Consistent beaconing activity observed
6/25/2025	11:40:46 AM	SYSTEM	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	Repeated POST to saltybug.com/update (status 200) site_proxy	File staging for exfiltration
6/25/2025	11:41:19 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\WindowsPowerShell\cmd.exe CommandLine c:\splunk-windows_sysmon	Sensitive file staging operation observed
6/25/2025	11:41:19 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - PowerShell copied random user file to C:\Windows\vs\secret1.txt Splunk-windows_sysmon	

Anti-Forensics and Evidence Deletion

Sysmon Event ID 1 shows execution of cmd.exe to delete artifacts. This occurred immediately after successful POST action, indicating:

- Exfiltration success
- Cleanup to reduce forensic traceability

Proxy logs and Sysmon confirm:

- Secrets.zip and secrets.txt were deleted post-exfiltration
- This qualifies as active anti-forensics behavior

6/25/2025	11:45:33 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\cmd.exe CommandLine c:\splunk-windows_sysmon	Cleanup - deleted secrets.txt
6/25/2025	11:45:33 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - cmd.exe executed 'cmd /c del C:\Windows\vs\secrets.txt' launched t Splunk-windows_sysmon	Data cleanup or artifact removal by malicious process
6/25/2025	11:46:05 AM	SYSTEM	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	HTTP POST request to C2 successful (200 OK) site_proxy	Persistent C2 communication detected
6/25/2025	11:46:05 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\cmd.exe CommandLine c:\splunk-windows_sysmon	Cleanup - deleted secrets.zip
6/25/2025	11:46:05 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - cmd.exe executed 'cmd /c del C:\Windows\vs\secrets.zip' launched t Splunk-windows_sysmon	Evidence deletion performed post exfiltration
6/25/2025	11:46:17 AM	-	acc-win10-13.site.lan	Sysmon Logs	PowerShell file deletion (secrets.txt, secrets.zip) Splunk-windows_sysmon	Anti-forensics / cleanup

Multi-Host Activity and Lateral Spread

Timeline correlation shows two hosts (ACC-WIN10-13 and ACC-WIN10-16) communicating with the same malicious C2 over a synchronized time window. Hostnames, user contexts, and artifact overlap confirm multi-host compromise, indicating either:

- Shared infection vector
- Phishing
- Common external download
- Internal spread

Cross-system correlation strengthens attribution and confirms elevated incident severity.

172	6/25/2025	11:45:33 AM	SYSTEM	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	POST request logged to saltybug.com/update	site_proxy	Network connection stable and active
172	6/25/2025	11:45:33 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate ExecutableInfo Image C:\Windows\System32\cmd.exe CommandLine c:\splunk-windows_sysmon		Cleanup - deleted secrets.txt
173	6/25/2025	11:45:33 AM	site\Administrator	acc-win10-13.site.lan	Sysmon Logs	Event ID: 1 ProcessCreate - cmd.exe executed 'cmd /c del C:\Windows\vs\secrets.txt' launched t Splunk-windows_sysmon		Data cleanup or artifact removal by malicious process
174	6/25/2025	11:46:05 AM	SYSTEM	ACC-WIN10-13 (172.16.6.113)	Proxy Logs	HTTP POST request to C2 successful (200 OK)	site_proxy	Persistent C2 communication detected

CONCLUSION

The activity observed across two Windows workstations is consistent with a controlled intrusion, data exfiltration, and active attacker persistence. The malware QRCodeScanner.exe was executed under elevated context, registered with a remote C2, collected local files, and transmitted them externally through encrypted HTTP POST traffic.

Evidence deletion and scheduled tasks confirm attacker attempts to maintain access and reduce forensic footprint. Activity correlation across hosts confirms expansion beyond a single endpoint compromise.

All forensic artifacts aligned across:

- Sysmon
- Proxy and PCAP logs
- Registry and LNK metadata
- Shellbag interactions
- Prefetch and MFT data

This creates a defensible, cross-validated chronology showing execution, C2, persistence, and cleanup.

RECOMMENDATIONS

1. Immediate containment:

- Isolate affected hosts from network
- Reset compromised credentials
- Block outgoing traffic to malicious domains
- Disable malicious scheduled tasks

2. Threat eradication:

- Remove QRCodeScanner.exe and related payloads
- Clear persistence mechanisms

- Confirm no additional tasks, scripts, or processes exist

3. **Infrastructure improvements:**

- Enforce application allowlisting
- Restrict administrative PowerShell execution
- Implement endpoint monitoring (Sysmon + EDR)
- Strengthen email/web filtering against drive-by links

4. **User awareness:**

- Educate on execution risks from external binary files
- Review phishing prevention posture

5. **Forensic retention:**

- Preserve full packet captures, logs, and forensic images
- Record artifact correlation for legal/IR documentation